Python Programming - 2301CS404

Lab - 7

Roll No : 315

Name : vishva M. Bhimani

01) WAP to find sum of tuple elements.

Sample Input:

```
T = (10, 20, 30, 40)
```

Sample Output:

```
Sum of tuple elements = 100
```

In [2]:
```
t1=(10,20,30,40)
sum=0
for i in range(len(t1)):
    sum=sum+t1[i]
print(sum)
```

```
100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

```
T = (7, 2, 9, 4, 1, 5)
K = 2
```

Sample Output:

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

In [8]:
```
t=(7,2,9,4,1,5)
k=int(input("enter value of k"))
l=sorted(t)
tuple(l)
min=l[:k]
max=l[-k:]
print(min,max)
```

```
[1, 2] [7, 9]
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

In [19]:
```
t = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
k=int(input("enter k"))
for i in t:
    for j in i:
        if(j%k!=0):
            break;
    else:
       print(i)
```

```
(2, 4, 6)
(4, 8, 12)
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27),
(4, 64), (5, 125)]
```

In [37]:
```
n=[1,2,3,4,5]
ans=[(i , i**3) for i in n]
print(ans)
```

```
[(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4,
-5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

In [47]:
```
t=[(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4, -5)]
s=[]
for i in t:
    for j in i:
        if(j<0 or j==0):
            break;
    else:
        s.append(i)
print(s)
```

```
[(1, 2, 3), (4, 5, 6)]
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

```
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8,
9)]
```

In [49]:
```
t=[(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
k=int(input("enter k"))
for i in t:
    if(len(i)==k):
        t.remove(i)
print(t)
```

```
[(3, 4, 5), (6,), (7, 8, 9)]
```

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

In [59]:
```
t= (3, 5, 3, 7, 3, 9)
l=list(t)

k=int(input("enter k"))
for i in l:
    if(i==k):
        l.remove(i)

print(tuple(l))
```

```
(5, 7, 9)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

In [63]:
```
t= (3, 5, 3, 7, 3, 9)
s=set(t)
s=tuple(s)
print(s)
```

```
(9, 3, 5, 7)
```

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

In [69]:

```
t = (2, 3, 4, 5)
l=list(t)
ans=[(l[i]*(l[i+1])) for i in range(len(l)-1)]
print(tuple(ans))
```

```
(6, 12, 20)
```

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

In [3]:

```
t1 = (1, 2, 3, 4)
t2 = (1, 2, 2, 3)

# for i in range(len(t1)):
#     for j in range(1,len(t1)):
#         if t1[i]==t1[j]:
#             print("false")
#             break
#     else:
#         print("true")
#         break

s1=set(t1)
s2=set(t2)
if(len(t2)==len(s2)):
    print("True")
else:
    print("False")
```

```
False
```

11) WAP to rotate the elements of a tuple to the right by `k` positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

In [4]:
```
T = (10, 20, 30, 40, 50)
k = int(input("enter k"))
n = len(T)
k = k % n

rotated_tuple = T[-k:] + T[:-k]

print(rotated_tuple)
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

In [8]:
```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
T=T1+T2

print(tuple(sorted(T)))
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

In [1]:
```
T = (4, 12, 7, 18, 10)
ans=tuple(T[i]**2 for i in range(0,len(T)) if T[i]%2==0 and T[i]>10)
print(ans)
```

```
(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

In [25]:
```
# n=[1,2,3,4,5]
# ans=[(i , i**3) for i in n]
# print(ans)
T = (4, 7, 9, 11, 15)
def is_prime(num):
    if num <= 1:
        return False
    for i in range(2, num):
        if num % i == 0:
            return False
    return True
ans=tuple((T.index(T[i]),T[i]) for i in range(0,len(T)) if is_prime(T[i]))
print(ans)
```

```
((1, 7), (3, 11))
```

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

In [29]:
```
T = ('a', 'b', 'c', 'd', 'e')
# ans1=tuple(T[i] for i in range(0,len(T)) if i%2!=0)
# ans2=tuple(T[i] for i in range(0,len(T)) if i%2==0)
# print("even=",ans2)
```

```
# print("odd=",ans1)
print("even=",T[::2])
print("even=",T[1::2])
```

```
even= ('a', 'c', 'e')
even= ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

In [48]:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])

for i in range(len(T[0])):
    s = 0
    for j in range(len(T)):
        s += T[j][i]
    print(s, end=" ")
```

```
12 15 18
```

In [ ]: